Ahmed Heakl

Ph.D. Student in Computer Vision, MBZUAI | Expected Graduation May 2030 (05/2030) | Abu Dhabi, UAE
Internship availability May to December 2027, full-time, 13+ weeks
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Research Interests

Vision-language models and multimodal reasoning, reinforcement learning and self-improving post-training for LLMs and VLMs, how models represent and update knowledge, efficient and adaptive inference, and multilingual document understanding.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) Aug. 2026 – present
Ph.D. in Computer Vision, expected graduation May 2030 *Abu Dhabi, UAE*

- Advised by Prof. Salman Khan and Prof. Fahad Shahbaz Khan, Intelligent Visual Analytics Lab (IVAL)

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) Aug. 2024 – May 2026
M.Sc. in Computer Vision, GPA 3.77/4.0 *Abu Dhabi, UAE*

- Master's Outstanding Research Award (2026). Coursework in computer vision, object detection, and VLMs

Egypt-Japan University of Science and Technology (E-JUST) Sep. 2019 – Jul. 2024
B.Sc. in Computer Science and Engineering, GPA 3.85/4.0, ranked 1st of 56 *Alexandria, Egypt*

- Theses on LLM-based code auto-grading and code-switched Arabic-English speech translation (ArzEn-LLM)

Research Experience

LG AI Research May 2026 – Sep. 2026
Research Scientist Intern *Seoul, South Korea*

- Led **VQS** (first author, under review at ICLR 2027), a self-evolution method for VLMs. A human study showed that 24% of majority-vote and 18% of model-judge labels in prior work are wrong, so VQS parses images into scene graphs, chart tables, or diagram graphs and computes exact answers with 70 program templates.
- Reached **94.4%** answer correctness in human evaluation (76.4% majority vote, 82.2% model judge) and improved Qwen3-VL by **+3.18**, **+2.32**, and **+2.43** points over 10 benchmarks at 2B, 4B, and 8B with GRPO, beating the best baseline at every scale and reaching **+3.84** after three cycles.
- Trained the image parser without any labels through best-of-K sampling scored by per-claim visual verification, raising parse precision from **80.8% to 87.6%**.

Parameter Lab, in collaboration with NAVER Jun. 2025 – May 2026
Research Scientist Intern *Tübingen, Germany*

- Authored **Dr.LLM** (ICLR 2026, first author), which retrofits frozen LLMs with lightweight per-layer routers that skip, execute, or repeat blocks, supervised by MCTS-derived layer paths and cutting the routing data requirement by **97%**.
- Improved accuracy by **+3.4%p** on ARC and DART while **saving 3–11 layers per query**, outperforming adaptive-depth baselines by up to **+7.7%** without touching base model weights.

Monta AI Jul. 2024 – Apr. 2025
Applied Scientist (remote) *California, USA*

- Trained a 7B vision-language model that converts diagrams and charts into executable Graphviz and Vega-Lite specifications, pretraining on 250k synthetic bilingual examples with **DDP across 8 GPUs on 2 nodes** and reaching 15.3% CER on held-out internal benchmarks.
- Co-led an Arabic OCR pipeline for a Saudi government client over 100k financial documents (layout, tables, text, and charts), with YOLOv10 reaching **95.3% mAP@0.5** on internal validation.

Earlier Experience

Software Engineer (remote), And Africa Co., Ltd., Tokyo, Japan Mar. 2023 – Jul. 2024
Built a Python route-optimization service (15+ objectives in under 2 s for 1,000 orders) and led the Logi-IQ platform.

Machine Learning Intern, Scoville Co., Ltd., Tokyo, Japan Jan. 2023 – Feb. 2023
Built a GAN-based authorship obfuscation model with GPT-2 and a Siamese ResNet (0.8 s per 1,024-token input).

Selected Publications

The Geometry of Forgetting: Temporal Knowledge Drift as an Independent Axis in LLM Representations

NeurIPS 2026 | Co-first author | arXiv:2605.09195

- Showed that temporal knowledge drift is a residual-stream direction orthogonal to correctness and uncertainty. A linear probe reaches **0.83–0.95 AUROC** across six models while entropy, CCS, and SAPLMA detectors stay near chance.

Program-Verified Self-Evolution for Vision-Language Models (VQS)

Under review, ICLR 2027 | First author | arXiv:2609.33855

- Programs over verified image parses write questions and compute exact answers for GRPO self-training, lifting label correctness to **94.4%** (vs. 76.4% majority vote) and improving Qwen3-VL by up to **+3.18** points over 10 benchmarks at 2B, 4B, and 8B.

Dr.LLM: Dynamic Layer Routing in LLMs

ICLR 2026 | First author | arXiv:2510.12773

- Per-layer routers let frozen LLMs skip, execute, or repeat blocks under MCTS supervision, giving **+3.4%p** accuracy with **3–11 fewer layers per query** and 97% less routing data.

DocAtlas: Multilingual Document Understanding Across 80+ Languages

EMNLP 2026 (Main) | First author | arXiv:2605.12623

- Model-free pipeline that renders 360k multilingual OCR pages across 82 languages with differential rendering and RTL synthesis, plus a 5.8k-page benchmark showing DPO gives stable cross-lingual transfer with minimal forgetting.

Additional Publications

- WorldCache: Content-Aware Caching for Accelerated Video World Models (*ECCV 2026*, second author)
- Mobile-O: Unified Multimodal Understanding and Generation on Mobile Devices (*Under review*, co-first author)
- VideoMolmo: Spatio-Temporal Grounding Meets Pointing (*ACL 2026 Findings*, co-first author)
- CASS: Nvidia to AMD Transpilation with Data, Models, and Benchmark (*ACL 2026 Main*, first author)
- SAHM: A Benchmark for Arabic Financial and Shari'ah-Compliant Reasoning (*ACL 2026 Main, Oral*, third author)
- MedRoundsQA: Persona-Aware Evaluation for Multi-Turn Medical Consultations (*EMNLP 2026 Main*, second author)
- Right Frame, Wrong Rule: Cultural Cues Expose the Financial Knowledge Gap They Were Meant to Close (*EMNLP 2026 Findings*, second author)
- CEPO: RLVR Self-Distillation using Contrastive Evidence Policy Optimization (*Under review*, first author)
- LlamaV-o1: Rethinking Step-by-step Visual Reasoning in LLMs (*ACL 2025 Findings*, co-first author)
- KITAB-Bench: Multi-Domain Arabic OCR and Document Understanding Bench (*ACL 2025 Findings*, co-first author)
- GG: CISC-to-RISC Transpilation with Testing Guarantees (*EMNLP 2025 Findings*, first author)
- SVRPBench: A Realistic Benchmark for the Stochastic Vehicle Routing Problem (*NeurIPS 2025 D&B*, co-first author)
- CAMEL-Bench: A Comprehensive Arabic Large Multimodal Model Benchmark (*NAACL 2025 Findings*, co-first author)
- AIN: The Arabic INclusive Large Multimodal Model (*Preprint*, first author)
- How Good are Foundation Models in Step-by-Step Embodied Reasoning? (*Under review*, second author)
- Earlier papers (2022–2024), first author on ArzEn-LLM and ResumeAtlas (*Procedia CS 2024*), AraSpider (*NLPBD 2024*), and music genre classification (*IJCNN 2022*), plus three applied vision papers (*ICINCO 2024*, *IEEE IMSA 2024*).

Honors, Awards, and Service

- Top Reviewer, NeurIPS 2026, and reviewer for ICLR 2026, NeurIPS 2026, and ICLR 2027 (vision-language papers)
- Master's Outstanding Research Award, MBZUAI, 2026. 3rd place nationally in ICPC, Egypt
- Fully funded M.Sc. scholarship (MBZUAI) and B.Sc. excellence scholarship (E-JUST)
- Fully funded exchange year at Kyushu University, Japan, selected from about 1,200 applicants

Open-Source Releases

AIN, Arabic-English multimodal models (2B to 72B), **1.5M+ downloads** | **CAMEL-Bench**, Arabic multimodal benchmark, 29k+ downloads | **LlamaV-o1**, visual reasoning model, 27k+ downloads | **Mobile-O**, on-device multimodal model shipped as an iOS app

Technical Skills

- **Programming** Python (primary), C/C++, CUDA, Java, Go, SQL, JavaScript
- **Deep learning** PyTorch, LLMs, VLMs, diffusion models and DiTs, multimodal and multilingual modeling, LoRA, RLHF, DPO, GRPO
- **Scale and tooling** multi-node DDP, FSDP, DeepSpeed, Slurm, inference profiling, Docker, CI/CD, AWS, GCP